

2024 ANNUAL REPORT

STATE OF DEVELOPER ECOSYSTEM

Language Trends, Satisfaction & Strategic Shifts

Based on survey of 23,262 developers worldwide

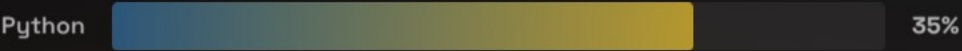


Python Closes the Gap: 8 Years of Language Usage Shifts

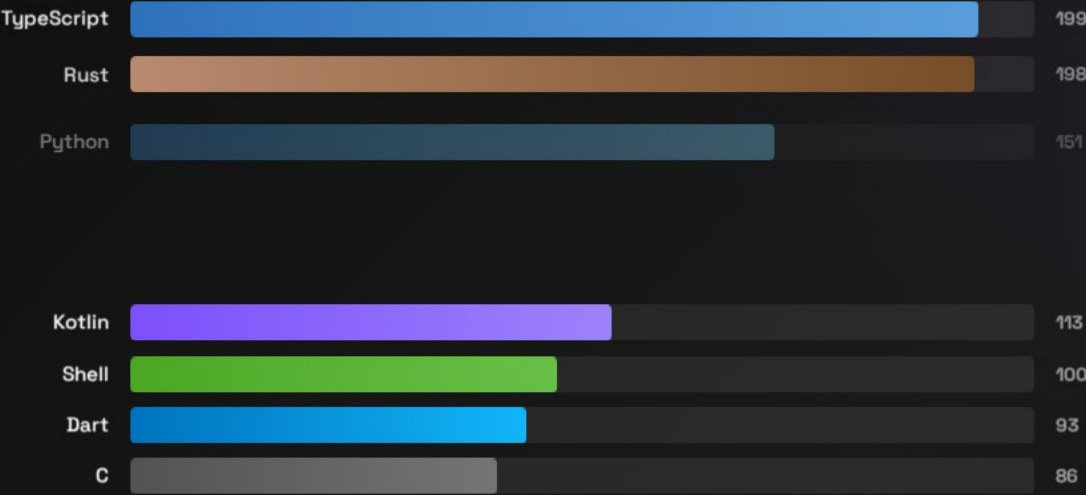
LANGUAGE	2017	2020	2022	2024	TREND
JavaScript	65%	70%	65%	61%	↓ Declining
Python	32%	55%	53%	57%	↑ Rising
HTML/CSS	60%	61%	54%	51%	↓ Declining
SQL	42%	56%	49%	48%	→ Plateaued
Java	47%	54%	48%	46%	↓ Slow decline
TypeScript	12%	28%	34%	37%	↑ Tripled
Go	8%	19%	19%	18%	→ Plateaued
Rust	—	7%	9%	11%	↑ Steady growth



Python Is Now the #1 Primary Language at 35%



TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198



Type-safe, modern-tooling languages dominate the top spots. Developer satisfaction correlates strongly with type safety and expressive type systems.



Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

AI / Machine Learning

31%

Python

Dominant in AI/ML due to libraries like TensorFlow, PyTorch, and scikit-learn



What This Means for Your Technology Strategy in 2025

01

Invest in Python for AI/ML

Python is the undisputed leader in artificial intelligence and machine learning. Teams working in AI should double down on Python expertise.

02

TypeScript is the Future of Frontend

TypeScript dominates UI development and has the highest developer satisfaction. It's the safest bet for frontend and full-stack investment.

03

Rust and Go Are Growth Plays

Both languages show rising usage and high satisfaction. For infrastructure and systems work, these are the languages to adopt.

04

PHP and Ruby Hiring Pools Shrinking

Both languages show long-term decline in usage. Consider migration paths and factor reduced talent availability into decisions.

05

Satisfaction Drives Retention

Type-safe, modern tooling languages correlate with higher developer satisfaction. Factor this into your technology stack choices.

